BOP GUARDIAN

Offshore BOP Monitoring Command Center

DATABRICKS APPS

Powered by Databricks Apps • Unity Catalog • Serverless

REAL-TIME DIGITAL TWIN & AGENTIC AI FOR BLOWOUT PREVENTER SAFETY

BOP Guardian delivers a unified command center that fuses live telemetry, predictive maintenance, SAP ERP integration, and multi-agent AI to protect the most safety-critical asset on any offshore drilling rig — the Blowout Preventer stack.

◆ BUSINESS VALUE

- **Reduce unplanned downtime** with AI-driven predictive maintenance and remaining useful life forecasting
- **Accelerate incident response** through automated severity-ranked recommendations and crew dispatch
- **Optimize spare parts spend** with real-time SAP ERP inventory visibility and min-stock alerts
- **Improve regulatory compliance** via continuous monitoring, event logging, and audit-ready dashboards
- **Centralize operations** — one platform for telemetry, maintenance, supply chain, and crew management

◆ CHALLENGES ADDRESSED

- **Siloed monitoring** — sensor data, work orders, and crew schedules live in disconnected systems
- **Reactive maintenance** — equipment failures detected only after they impact operations
- **Slow decision-making** — manual anomaly analysis delays critical safety responses
- **Certification blind spots** — no unified view of crew qualifications for BOP interventions
- **Data latency** — batch pipelines miss time-sensitive pressure and temperature excursions

◆ WHO IT'S FOR

🛢️ **Drilling Engineer**
Real-time BOP health visibility and anomaly alerts during drilling operations

🔧 **Maintenance Manager**
Predictive RUL forecasting, work order tracking, and spare parts optimization

🛡️ **HSE / Safety Officer**
Continuous compliance monitoring, event audit trails, and severity dashboards

📊 **Operations Director**
Fleet-wide KPI visibility, cost optimization, and strategic planning insights

10	5	8	3s
BOP Components Monitored	AI Sub-Agents	Dashboard Tabs	Live Refresh Interval

◆ QUICK START GUIDE

1. **Clone Repository**
   Clone the BOP Guardian repo to your local machine or workspace
2. **Import to Workspace**
   Use Databricks CLI to import app files into your workspace
3. **Deploy the App**
   Run `databricks apps deploy` to launch the command center
4. **Monitor & Explore**
   Open the app URL, explore all 8 tabs, and interact with Guardian AI

◆ GETTING THE MOST OUT OF BOP GUARDIAN

📡 **Watch the Digital Twin**
Use the BOP Status tab to monitor live P&ID health indicators and pressure readings in real time

🤖 **Ask Guardian AI**
Chat with the Guardian Advisor for natural-language insights, recommendations, and root-cause analysis

⚙️ **Track Predictive Maintenance**
Review RUL predictions and failure probabilities to plan interventions before breakdowns occur

👥 **Manage Crew & Work Orders**
Use Crew & Ops and SAP ERP tabs to dispatch certified personnel and track spare parts inventory

◆ TARGET INDUSTRY

Upstream Oil & Gas — Offshore drilling operations, deepwater rigs, subsea BOP systems, and FPSO vessels. Applicable to any operator managing high-pressure well control equipment under API 53 / BSEE regulations.

⚠ **DEMO ONLY** — This application is a demonstration Databricks App. Please refer to the included LICENSE file for full terms of use.

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